

Fatal Pedestrian & Cyclist Crash Risk Factors in Oregon

A Spatial & Predictive Analysis (2015-2024)

Simon Thompson · Rohan Srinivasa Babu

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Abstract

Fatal pedestrian and cyclist crashes are not distributed evenly across Oregon. Using 2015-2024 FARS crash records enriched with Census data, this project identifies geographic concentrations and examines the roadway, environmental, temporal, and community characteristics associated with those fatalities. The goal is to help transportation agencies prioritize where further safety investigation and investment are most needed.



Introduction & Motivation

Problem & question

Pedestrians and cyclists are physically vulnerable in roadway collisions, and fatal crashes may reflect recurring patterns in road design, lighting, speed, location, and surrounding community conditions. We ask which environmental, roadway, and temporal factors are most strongly associated with fatal pedestrian and cyclist crashes in Oregon from 2015-2024, and whether these crashes are disproportionately concentrated in communities with particular demographic or socioeconomic characteristics.

Impact & stakeholders

The findings may support ODOT, city and county planners, public-health organizations, and community advocates in identifying locations and conditions that warrant closer safety review. Pedestrians and cyclists are the primary affected stakeholders, while transportation agencies are positioned to act on the results.



Background & Related Work

FARS provides standardized national records for police-reported motor-vehicle crashes involving at least one fatality. Transportation-safety studies commonly combine crash records with spatial hotspot analysis, roadway characteristics, and statistical modeling to identify high-risk locations and contributing conditions.

This project adds neighborhood context by spatially joining fatal crash locations to Census tracts and ACS indicators. It differs from a purely descriptive crash map by combining spatial clustering, statistical testing, predictive modeling, and an equity-focused analysis within one statewide Oregon dataset.



Reproducibility & References

Repository – <https://github.com/S1mT/data510-OregonPedSafety> · Key dependencies – Python 3.x, pandas, Geopandas, shapely, numpy, scikit-learn, seaborn, matplotlib, requests
References: [1] Author, Title, Venue, Year. [2] ...



Data, Engineering & Ethics

Sources & scale

The final dataset contains **1,049 fatal pedestrian and cyclist victim records**, representing **490 unique Oregon crashes** from 2015–2024. It combines NHTSA FARS accident, person, pedestrian/bicyclist, vehicle, and weather tables with Census tract boundaries and ACS demographic indicators. The final analysis-ready dataset contains 38 variables.

Pipeline

FARS annual CSVs (2015-2024) → filter Oregon records (STATE = 41) and standardize schemas → create annual accident, person, ptype, vehicle, and weather tables → join victim, crash, weather, and striking-vehicle records using ST_CASE, VEH_NO, PER_NO, and STR_VEH → concatenate annual outputs into FARSinfrastructure.csv → validate coordinates and convert crash points to a GeoDataFrame → spatially join points to Oregon Census tracts → query 2022 ACS 5-year demographic and commuting variables → calculate tract-level percentages and merge on GEOID → export the 38-variable analysis-ready FARSmaster.csv.

Ethics & responsible use

The project uses public, anonymized government data with no direct personal identifiers. FARS includes only crashes involving at least one fatality, so findings describe fatal-crash patterns rather than total pedestrian or cyclist risk. Census variables describe neighborhoods, not individual victims, and associations should not be interpreted as causation. Results should guide equitable investigation and investment rather than stigmatize communities identified as hotspots.



Methods & Evaluation

Approach

Spatial Clustering (DBSCAN): We applied Density-Based Spatial Clustering of Applications with Noise (DBSCAN) using a 200m radius and a minimum threshold of 2 fatalities. This algorithm is ideal because it identifies localized, high-density crash hotspots based on actual road networks, filtering out isolated and rare incidents.

Infrastructure Profiling: For each hotspot, we calculated the statistical mode (most frequent occurrence) across key FARS variables (e.g., speed limit, road type, lighting) to build a "Systemic DNA" profile of high-risk design flaws.

Demographic Correlation: We calculated Pearson correlation coefficients to measure the linear relationship between Census tract demographics (e.g., median income, % Black population) and the tract-level fatal crash rate.

Validation

Spatial Validation: We cross-referenced our mathematically derived hotspots against satellite imagery and ODOT crash logs to confirm the presence of high-risk infrastructure (e.g., multi-lane arterials, mid-block crossings).



Results & Visualizations

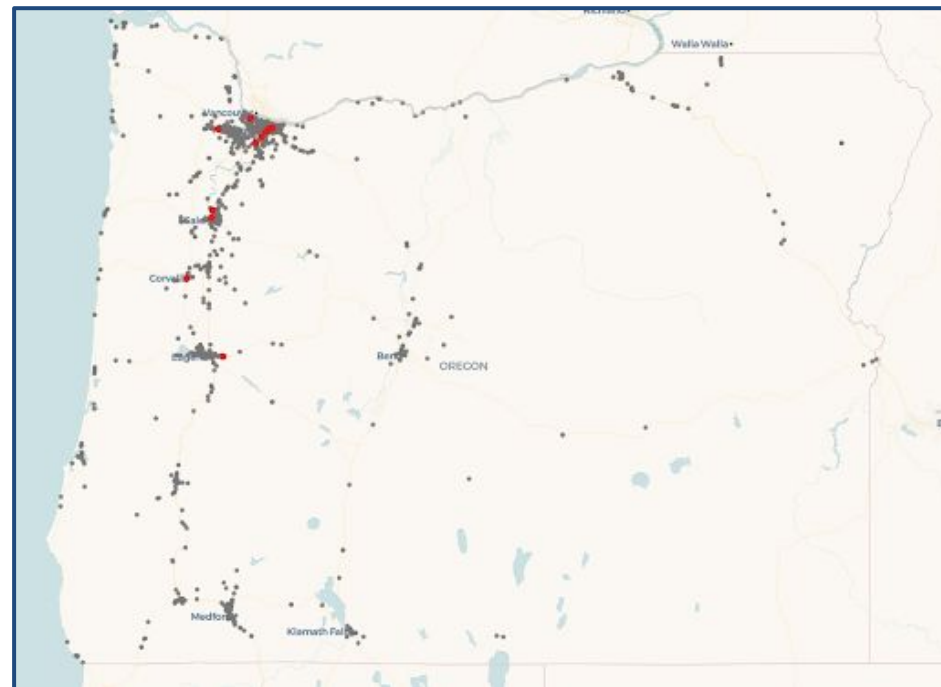


Fig. 1: All Crash Records Visualized With Identified Clusters

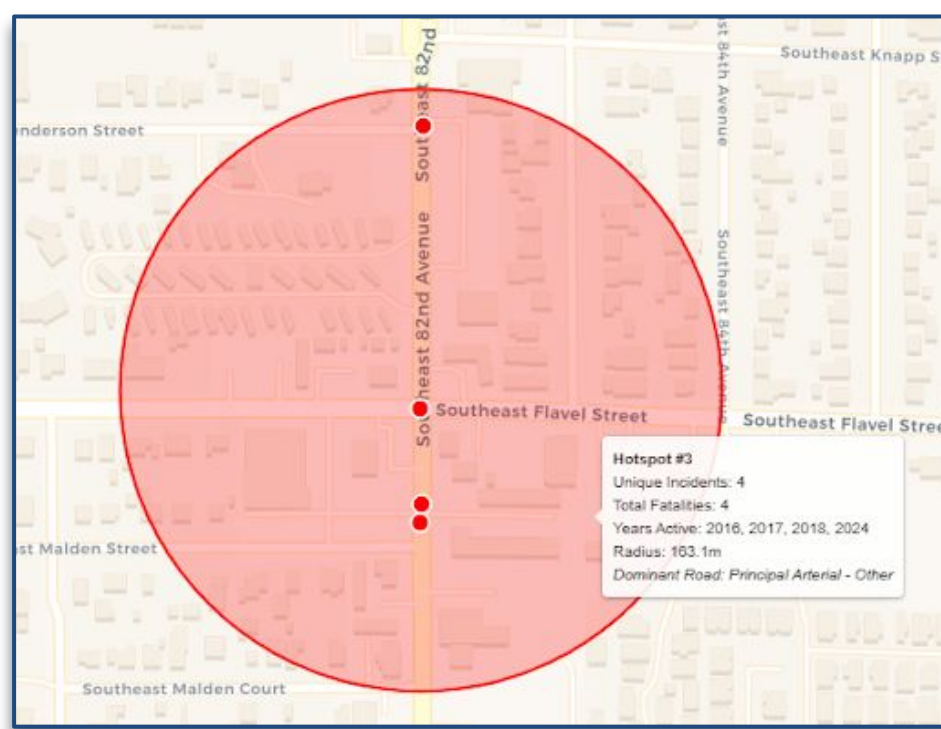


Fig. 2: Cluster #1 - Southeast 82nd Ave, Portland OR

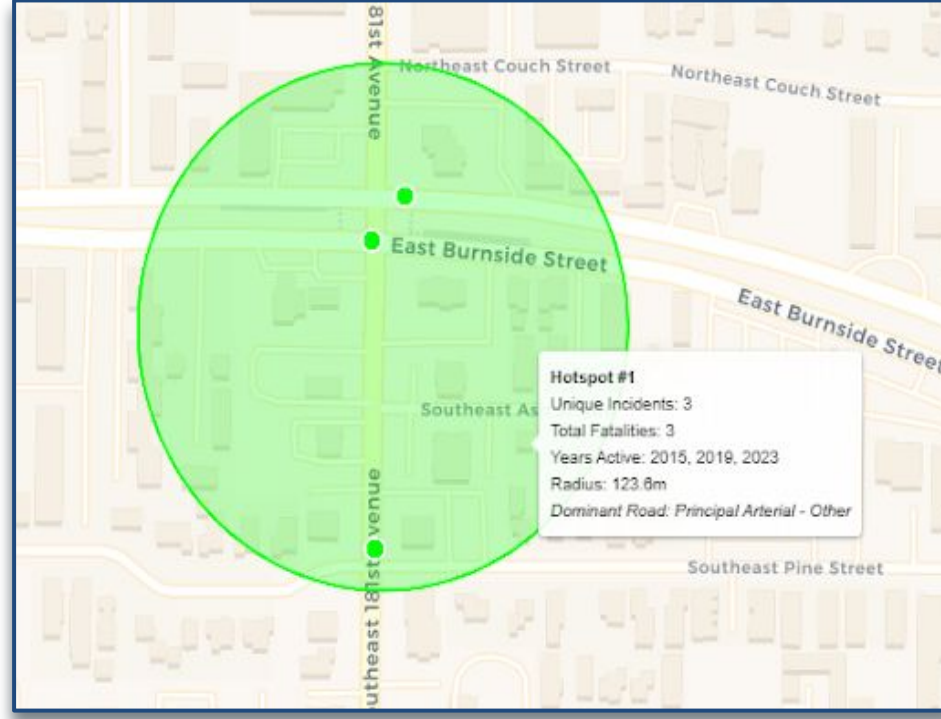


Fig. 3: Cluster #2 - Southeast 181st Avenue, Portland, OR

Feature	Percentage Occurrence
Arterial Road	92.5%
Not an Intersection	57.5%
Two-Way With a Continuous Left-Turn Lane	50.0%

Fig. 7: Cluster Feature Analysis

crash_rate_per_10k	1.000000
MEDIAN_INCOME	-0.135552
PCT_WHITE	-0.077671
PCT_BLACK	0.202264
PCT_HISPANIC	0.002306
PCT_TRANSIT_WALK_COMMUTE	0.184905

Fig. 4: Correlation with Crash Rate per 10k Population

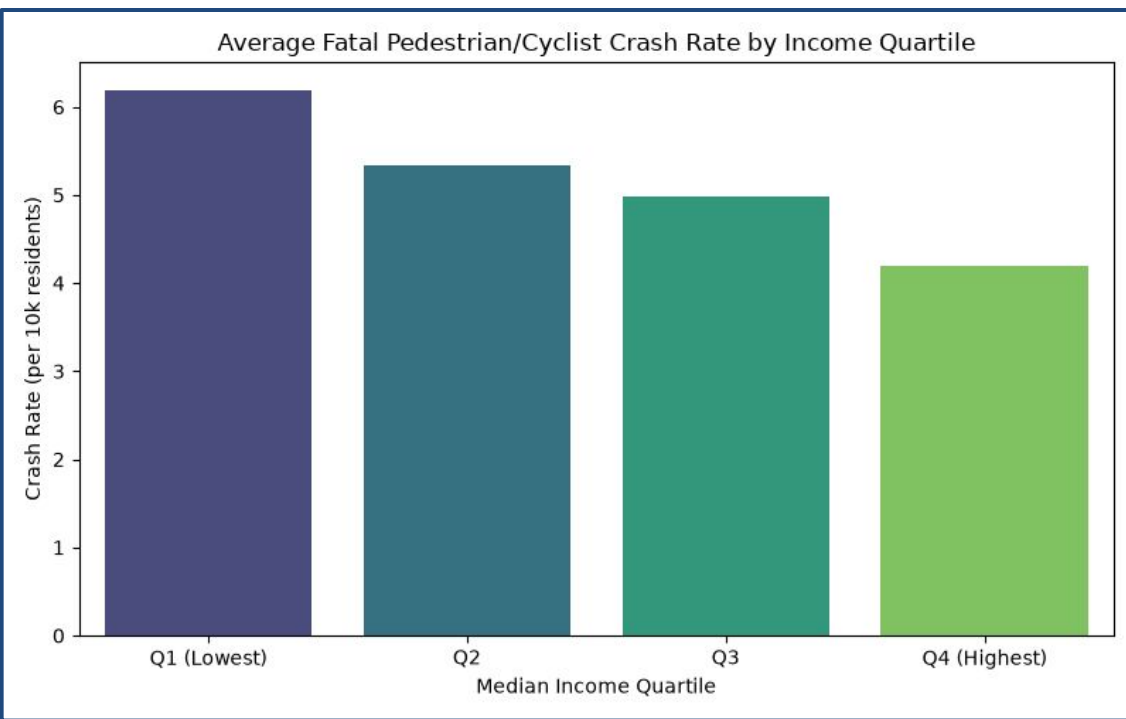


Fig. 5: Average Fatal Crash Rate by Income

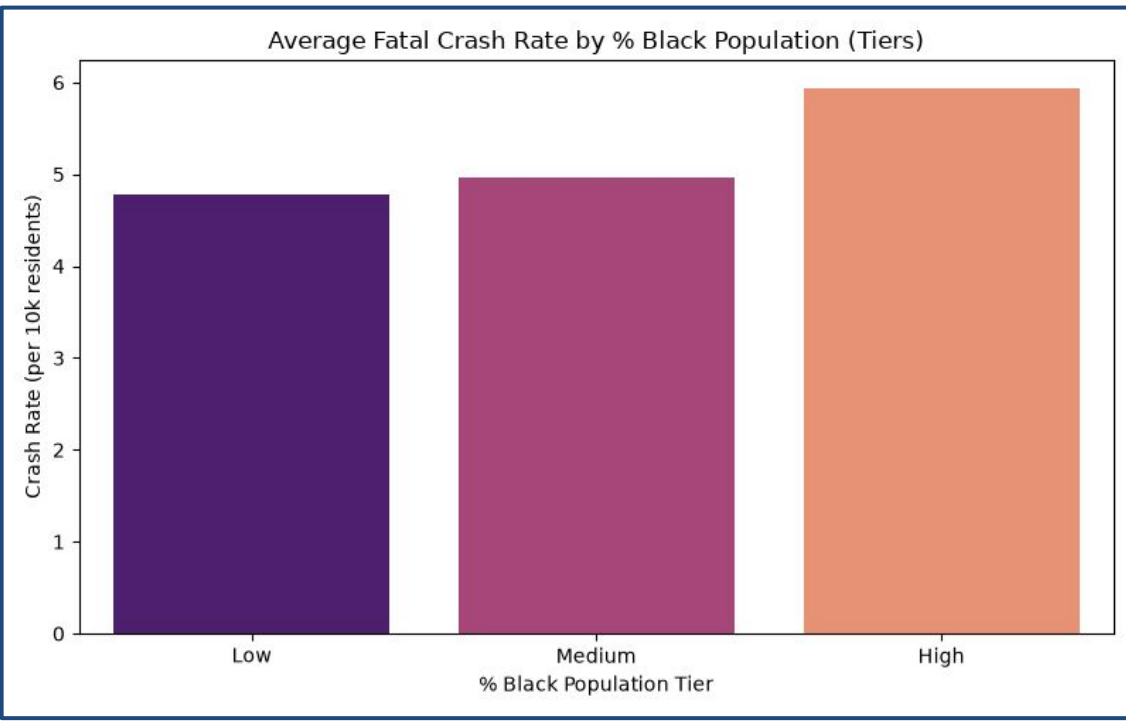


Fig. 6: Average Fatal Crash Rate by Black Population

Feature	Percentage Occurrence
Arterial Road	60.4%
Not an Intersection	70.6%
Two-Way With a Continuous Left-Turn Lane	23.7%

Fig. 8: Statewide Feature Analysis



Discussion, Limitations & Conclusions

Discussion

Patterns supported across maps, descriptive statistics, and predictive models provide stronger evidence than any single method. Hotspots should guide further local review, not be treated as proof of causation.

Limitations & future work

FARS excludes non-fatal crashes and lacks pedestrian, cyclist, and traffic exposure measures. Missing roadway data also limits interpretation. Future work should add non-fatal crashes, exposure counts, updated Census data, and detailed infrastructure records.

Conclusion

This project identifies where fatal pedestrian and cyclist crashes concentrate and which factors are associated with them. The results can help Oregon agencies prioritize corridors and communities for further analysis and targeted safety investment.

